



## Risk Classification in LambdaProof

### What Is Risk Classification?

Every action an AI agent might take carries different levels of potential impact. Browsing a public website is fundamentally different from transferring funds or sending emails on your behalf. LambdaProof's Risk Classifier evaluates each proposed action and assigns it to a risk class, which determines what level of proof and authorization is required before the action can proceed.

The classifier itself is proprietary—it represents one of LambdaProof's core innovations in balancing safety with usability. However, the *output* (the risk classes and what they mean) is fully transparent.

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### Risk Classes

LambdaProof defines four risk classes:

| Risk Class             | Description                                | Proof Requirement | Example Actions                                                |
|------------------------|--------------------------------------------|-------------------|----------------------------------------------------------------|
| <b>R0 - Negligible</b> | Actions with no meaningful external impact | None              | Reading public data, local computation, displaying information |

|                      |                                                                                           |                                              |                                                                                  |
|----------------------|-------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------|
| <b>R1 - Low</b>      | Reversible actions with limited scope                                                     | Session attestation                          | Navigating websites, saving drafts, querying non-sensitive APIs                  |
| <b>R2 - Moderate</b> | Actions with real-world consequences that may be difficult to reverse                     | Capability token + evidence binding          | Sending emails, posting to social media, modifying files                         |
| <b>R3 - High</b>     | Irreversible actions or actions with significant financial, legal, or safety implications | Full ZK proof + explicit human authorization | Financial transfers, signing contracts, accessing medical records, deleting data |

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## How Risk Class Affects Authorization

The risk class determines the *obligations* the system imposes before allowing an action:

**R0 (Negligible):** Action proceeds with minimal logging. No token required.

**R1 (Low):** Action requires a valid session. The system verifies the agent is operating within an authenticated context and logs the intent.

**R2 (Moderate):** Action requires a capability token bound to the specific intent. The token encodes what action is permitted, for how long, and under what constraints. Evidence of user consent or delegation must be referenced.

**R3 (High):** Action requires cryptographic proof that all preconditions are satisfied—including explicit human authorization for that specific action. The proof is verifiable on-chain or via the Archivum audit layer. These actions are never auto-approved.

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## What the Classifier Considers

While the classification algorithm is proprietary, the *factors* it considers are public:

- **Reversibility** — Can the action be undone?
- **Scope** — How many systems, people, or resources are affected?

- **Financial exposure** — Does this involve money or assets?
- **Data sensitivity** — Does this touch personal, medical, or confidential information?
- **Authorization chain** — Was this action explicitly requested, delegated, or inferred?
- **Temporal pressure** — Is there urgency that might bypass deliberation?
- **Recipient trust** — Is the target a known entity or unknown?

These factors are weighted and combined. The weights and combination logic are trade secrets—this is where AProof's competitive advantage lies.

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## Why This Matters

Risk classification enables *proportional* security:

- **Low-friction for low-risk:** Agents can operate fluidly for routine tasks
- **High-friction for high-risk:** Critical actions require explicit proof of authorization
- **Auditability:** Every action's risk class is logged, enabling post-hoc review
- **Adaptability:** The classifier can be tuned per deployment context (enterprise, consumer, healthcare, etc.)

Without risk classification, you'd face a binary choice: either lock down everything (unusable) or permit everything (unsafe). AProof's approach enables the middle path—*calibrated trust*.

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## What's Public vs. Private

| Aspect                                  | Visibility                     |
|-----------------------------------------|--------------------------------|
| Risk class definitions (R0-R3)          | ✓ Public                       |
| Factors considered in classification    | ✓ Public                       |
| How factors are weighted/combined       | ✗ Proprietary                  |
| Classification algorithm implementation | ✗ Proprietary                  |
| Per-deployment tuning parameters        | ✗ Proprietary                  |
| Audit logs of classification decisions  | ✓ Available to data controller |

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## For Builders

If you're building on  $\Lambda$ Proof, you interact with risk classification through the Policy Evaluation API:

```
POST /policy/evaluate
{
  "intent_hash": "a1b2c3d4..."
}
```

Response:

```
{
  "decision": "ALLOW",
  "risk_class": "R2",
  "proof_level": "ATTESTATION",
  "obligations": ["bind_evidence", "log_audit_event"],
  "token_ttl": 300
}
```

You receive the *result* of classification, not the reasoning. This is intentional—it prevents gaming the classifier while still giving you everything you need to build compliant integrations.

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## Questions We Expect

### "Can I override the risk class?"

No. Risk classification is determined by the system based on the action's properties. However, policy bundles can specify *additional* constraints on top of the base classification.

### "What if I disagree with a classification?"

The classification factors are public. If you believe an action is misclassified, you can submit feedback. Systematic misclassifications are reviewed and the model is updated—but the update process is internal.

### **"Can I see why an action was classified a certain way?"**

Audit logs include the risk class and the factors that were present, but not the weights or decision logic. This provides accountability without exposing the algorithm.

### **"Is the classifier an ML model?"**

We don't disclose implementation details. The classifier produces consistent, auditable results regardless of its internal architecture.

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Let me know if you want me to adjust the tone (more technical, more accessible), add sections (e.g., "For Regulators," "For Security Auditors"), or tighten the scope. This draft tries to be maximally transparent about *what* and *why* while protecting *how*.

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1. Milestone-Vision-Timeframe.csv
    - a. Aproof Healthcare Protocols V0.1.pdf
    - b. Adapps On Web4 - Developer Overview.pdf
  2. Building a  $\Lambda$ Dapp\_ A Developer's Implementation Guide.pdf
  3. MTPI\_Instructions.pdf
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  5. MTPI\_RootContract\_Project.pdf
  6. Patent Search Results.pdf
  7. The  $\Lambda$ Proof Architecture\_ A Comprehensive Analysis of its Implications for Web4.pdf
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